

## CSC 3114 Cryptology and Coding Theory

### Revision Question Bank 1

**Conventions:** Use the English alphabet A–Z. For letter ciphers, ignore spaces and punctuation unless otherwise stated. Caesar encryption shifts letters forward, with wraparound. For Vigènere calculations, use  $A = 0, B = 1, \dots, Z = 25$  and addition modulo 26. One byte is 8 bits; 1 KiB is 1024 bytes.

### A. Cryptography and Classical Ciphers

#### Question 1: Foundations and security goals [10 marks]

- (a) Distinguish cryptography, cryptanalysis and cryptology. [3 marks]
- (b) Define plaintext, ciphertext and a cryptographic key. [3 marks]
- (c) Explain confidentiality and integrity. Give one example of a violation of each. [4 marks]

#### Question 2: Encryption and adversaries [10 marks]

- (a) Draw a labelled encryption–decryption model for Alice sending a message to Bob over an insecure channel. Include the key, plaintext, ciphertext and Eve. [4 marks]
- (b) Distinguish symmetric encryption from asymmetric encryption in terms of the keys used and which keys must remain secret. [3 marks]
- (c) Eve modifies Alice’s message and sends it to Bob while pretending to be Alice. Identify two security goals threatened. [3 marks]

#### Question 3: Caesar and ROT13 ciphers [10 marks]

- (a) Encrypt `ATTACK AT ONCE` using a Caesar shift of 4. [3 marks]
- (b) Decrypt `VHFUHW PHVVDJH` using a Caesar shift of 3. [3 marks]
- (c) Explain why applying ROT13 twice recovers the original message. [2 marks]
- (d) Explain why a Caesar cipher is vulnerable to brute-force cryptanalysis, stating the number of non-identity shifts. [2 marks]

#### Question 4: Substitution techniques [10 marks]

- (a) Distinguish substitution from transposition. [2 marks]
- (b) Compare monoalphabetic and polyalphabetic substitution, using Caesar and Vigènere as examples. [4 marks]
- (c) Explain homophonic substitution and polygram substitution. Give a simple illustrative mapping for each. [4 marks]

## A. Cryptography and Classical Ciphers (continued)

### Question 5: Transposition ciphers

[10 marks]

- (a) Encrypt **HELLO WORLD** using the two-row rail fence method: place successive letters alternately in the upper and lower rows, then read the upper row followed by the lower row. Show the arrangement. [3 marks]
- (b) Encrypt **DEFEND THE EAST WALL OF THE CASTLE** using regular columnar transposition with the keyword **GERMAN**. Write the letters row by row, pad with **X** to complete the rectangle, and read columns from top to bottom in alphabetical order of the keyword letters. Show the rectangle and the ciphertext. [5 marks]
- (c) Distinguish regular from irregular columnar transposition. [2 marks]

### Question 6: The Vigenere cipher

[10 marks]

- (a) Explain how the keyword is aligned with a message longer than the keyword. [2 marks]
- (b) Encrypt **COMPUTING GIVES INSIGHT** with the keyword **LUCKY**. Show the repeated key and letter-by-letter working. [5 marks]
- (c) Use the same key to decrypt the first five ciphertext letters obtained in part (b), showing your working. [3 marks]

### Question 7: XOR and the Vernam cipher

[10 marks]

- (a) Write the truth table for XOR and calculate  $0101 \oplus 0011$  and  $0010 \oplus 1010$ . [3 marks]
- (b) Encrypt **DOG** with the key **CAT** using bitwise XOR. Use the following 8-bit character codes and give the ciphertext as three binary bytes. [4 marks]

| Character | D        | O        | G        | Character | C        | A        | T        |
|-----------|----------|----------|----------|-----------|----------|----------|----------|
| Code      | 01000100 | 01001111 | 01000111 | Code      | 01000011 | 01000001 | 01010100 |

- (c) Recover the first plaintext byte by applying XOR again with the corresponding key byte. Explain why Vernam ciphertext is often displayed as binary, decimal or hexadecimal values. [3 marks]

### Question 8: One-time pads and cipher operation

[10 marks]

- (a) State four requirements concerning key length, randomness, reuse and secrecy for a secure one-time pad. [4 marks]
- (b) Explain how a one-time pad differs from a Vigenere cipher that repeats a short keyword, and why randomness matters. [3 marks]
- (c) Distinguish a bit-stream cipher from a block cipher in terms of how the plaintext is processed. Name two operations that may be used in a block cipher. [3 marks]

## B. Introduction to Coding Theory

### Question 9: Purpose and applications of coding theory [10 marks]

- (a) Define coding theory and distinguish its main purpose from that of cryptography. [3 marks]
- (b) Explain why errors can arise both when information is transmitted and when it is stored for later retrieval. [3 marks]
- (c) Identify two applications of error-correcting codes and explain the benefit in each case. [4 marks]

### Question 10: The communication system [10 marks]

- (a) Draw a labelled digital communication model showing the source, source encoder, channel encoder, modulator, noisy channel, demodulator, channel decoder, source decoder and sink, in the correct order. [5 marks]
- (b) Distinguish source coding from channel coding. [3 marks]
- (c) Explain the roles of the modulator and demodulator. [2 marks]

### Question 11: Error detection and error correction [10 marks]

- (a) Explain how redundancy can help a receiver detect or correct transmission errors. [3 marks]
- (b) Distinguish backward error correction (BEC) from forward error correction (FEC), including what the receiver does after an error is detected. [4 marks]
- (c) Explain the basic idea of a parity bit and why detecting an error does not necessarily identify which bit is wrong. [3 marks]

### Question 12: A three-bit error-detecting code [10 marks]

The source codes are apple  $\rightarrow$  00, banana  $\rightarrow$  01, cherry  $\rightarrow$  10, grape  $\rightarrow$  11. The channel encoder uses

$$00 \rightarrow 000, \quad 01 \rightarrow 011, \quad 10 \rightarrow 101, \quad 11 \rightarrow 110.$$

- (a) Explain why receiving 01 after transmitting the source word 00, without channel encoding, need not reveal that an error occurred. [2 marks]
- (b) List all words obtainable from 000 by exactly one bit error. Explain how the channel code detects each error. [3 marks]
- (c) If 100 is received and exactly one bit error occurred, list every possible transmitted codeword. Explain why correction is ambiguous. [3 marks]
- (d) Calculate the code rate and state the number of redundant bits per encoded message. [2 marks]

**B. Introduction to Coding Theory (continued)****Question 13: A five-bit error-correcting code****[10 marks]**

Consider the channel encoding scheme

$$00 \rightarrow 00000, \quad 01 \rightarrow 01111, \quad 10 \rightarrow 10110, \quad 11 \rightarrow 11001.$$

- (a) List all five received words resulting from exactly one bit error in 00000. **[2 marks]**
- (b) For the received word 10000, count the differing bit positions relative to each valid codeword. Assuming at most one error, identify the transmitted codeword and recover the two-bit source word. **[4 marks]**
- (c) Determine the minimum distance by comparing the valid codewords. Show the pairwise counts. **[3 marks]**
- (d) Calculate the code rate. **[1 marks]**

**Question 14: Repetition coding and majority decoding****[10 marks]**

A two-bit source word is repeated five times, as in  $01 \rightarrow 0101010101$ .

- (a) Encode the source word 10. **[2 marks]**
- (b) Decode 1100100010 by applying majority voting separately to odd-numbered and even-numbered positions. Show both groups and the recovered word. **[4 marks]**
- (c) State how many bit errors this scheme is guaranteed to correct and justify the answer using majority voting. **[2 marks]**
- (d) Calculate its code rate and explain one disadvantage of repetition coding. **[2 marks]**

**Question 15: Redundancy and transmission efficiency****[10 marks]**

- (a) A source word contains  $k$  bits and is repeated  $2r + 1$  times. Express the transmitted length and code rate in terms of  $k$  and  $r$ . **[3 marks]**
- (b) For  $k = 2$  and  $r = 3$ , determine the transmitted length, number of redundant bits and guaranteed number of correctable bit errors. **[3 marks]**
- (c) Compare the rates of the two-to-three-bit code, the two-to-five-bit code and fivefold repetition of a two-bit word. Explain the trade-off illustrated by these examples. **[4 marks]**

**Question 16: Channel coding concepts and design goals****[10 marks]**

- (a) Define minimum distance and error-correction capability. **[2 marks]**
- (b) Explain channel capacity and its significance for communication over a noisy channel. **[2 marks]**
- (c) State four goals to consider when designing channel encoders and decoders. **[4 marks]**
- (d) Describe two practical benefits of improved error-correcting codes. **[2 marks]**

### C. Steganography

**Question 17: Steganography, cryptography and watermarking** [10 marks]

- (a) Define steganography and compare its primary goal and visible output with those of cryptography. [3 marks]
- (b) Distinguish steganography from digital watermarking in terms of purpose and the relationship between the hidden information and its carrier. Give one example of each. [4 marks]
- (c) Explain the separate roles of encryption, embedding and authentication when used together to protect a hidden message. [3 marks]

**Question 18: Embedding and extraction** [10 marks]

- (a) Define cover object, payload, stego-key and stego-object. [4 marks]
- (b) Draw a labelled embedding and extraction model. Write equations for producing the stego-object and recovering the payload, and state the condition for correct recovery. [4 marks]
- (c) Explain why relying only on secrecy of the embedding program is weaker than using a secret stego-key. [2 marks]

**Question 19: Historical and text-based techniques** [10 marks]

- (a) Describe any two historical steganographic methods. Explain how each conceals information. [4 marks]
- (b) Recover the hidden message by taking the first letter of each word in the following sentence: [2 marks]

Since Everyone Can Read, Encoding Text In Neutral Sentences Is Doubtfully Effective.

- (c) Suppose one space encodes 0 and two spaces encode 1. Represent 10110 as a sequence of five gap widths. State three additional details the receiver needs for unambiguous extraction. [4 marks]

**Question 20: Least-significant-bit image embedding** [10 marks]

An RGB pixel has component values (201, 118, 45), represented by

$$R = 11001001, \quad G = 01110110, \quad B = 00101101.$$

- (a) Explain the principle of least-significant-bit (LSB) embedding and why small changes to image components may be difficult to notice. [3 marks]
- (b) Embed the payload bits 010 by replacing the LSB of R, G and B respectively. Show the resulting binary values and decimal RGB triple. [4 marks]
- (c) Extract the three payload bits from your modified pixel. Explain why a visually ordinary image is not necessarily free of hidden data. [3 marks]

### C. Steganography (continued)

**Question 21: Image capacity and file formats** [10 marks]

- (a) Write the raw image-payload capacity formula in terms of width, height, number of colour channels and embedded bits per channel. Define the units of the result. [2 marks]
- (b) Calculate the raw capacity of a  $1024 \times 768$  RGB image when one LSB per channel is used. Give your answer in bits, bytes and KiB. [3 marks]
- (c) State two reasons why usable payload capacity may be smaller than raw capacity. [2 marks]
- (d) Explain why the image format matters to payload survival. Compare PNG pixel preservation with JPEG quantization, and identify a risk of embedding through GIF palette indices. [3 marks]

**Question 22: Audio and other embedding approaches** [10 marks]

- (a) Briefly explain LSB coding, echo hiding, phase coding and spread spectrum in audio steganography. [4 marks]
- (b) Explain how resampling, noise reduction and lossy compression can affect hidden audio payloads. [3 marks]
- (c) Explain the purpose of a mimic function in steganography and how changing a message's statistical profile may help it avoid automated attention. [3 marks]

**Question 23: Carriers, applications and channel risks** [10 marks]

- (a) Identify two places where a payload may be embedded in video and two features that can carry hidden information in network communication. [4 marks]
- (b) Explain how video transcoding and network normalization may disrupt these hidden channels. [2 marks]
- (c) Give two legitimate applications and two potential misuses of steganography. [4 marks]

**Question 24: Steganalysis and detection errors** [10 marks]

- (a) Define steganalysis. [2 marks]
- (b) Describe structural, visual or auditory, statistical, and machine-learning approaches to steganalysis. [4 marks]
- (c) A detector flags a clean image as containing hidden data and fails to flag another image that does contain hidden data. Name the two types of error and explain each. [2 marks]
- (d) Explain why failure to notice visible or audible changes alone is insufficient evidence that no information has been hidden. [2 marks]